READING PASSAGE 1

*You should spend about 20 minutes on **Questions 1–13**, which are based on Reading Passage 1 below.*

The History of the Pencil

Pencils have existed for millennia. It is known that the ancient Egyptians, Greeks and Romans all used small pieces of metal as writing instruments, and that 14th-century European artists made pale grey drawings with thin rods of lead, silver or zinc. It is, however, generally accepted that the history of the modern pencil began in the 16th century, in the hilly, rural county of Cumberland in north-west England.

Some time before 1565, an enormous deposit of the soft, dark grey mineral now known as graphite was discovered in the Borrowdale district of the county. Then, as now, Cumberland was a farming area, and the locals found the mineral useful for marking their sheep so they could identify the ones that belonged to them. It was thought to be a form of lead, and so was referred to as *plumbago*, a term deriving from the Latin name for that element. However, the mineral was later identified as a form of carbon, and in 1789 the German geologist Abraham Gottlob Werner renamed it *graphite*, a word based on the ancient Greek *grapho*, meaning ‘draw’ or ‘write’.

The graphite found at Borrowdale was extremely dense and it could easily be sawn into sticks to be used for marking, writing or drawing. But because graphite is also soft, it requires some form of encasement around it, and so the sticks were initially wrapped in string for stability and ease of use. The news of the usefulness of these early pencils spread far and wide, attracting the attention of artists throughout Europe.

In England, the government soon realised the value of the Borrowdale find, particularly for national defence purposes, because graphite proved ideal for lining the moulds used in the making of cannon balls. The mines were therefore taken over by the government, which guarded and periodically flooded them to prevent theft. Because of this, there were occasions when supplies for use in pencils had to be smuggled out in secret. Nevertheless, England still became the centre of a thriving pencil-making industry.

Although deposits of graphite existed in other parts of the world, they were not of the same exceptional purity and quality as Borrowdale graphite, and had to be crushed to a powder to get rid of the impurities. England therefore continued to enjoy a monopoly on the production of pencils until a method of reconstituting the graphite powder could be found. English pencils continued to be made with sticks cut from pure Borrowdale graphite into the 1860s, after which the design was modified. The town of Keswick, close to the Cumberland graphite find, still manufactures pencils today, the factory also being the location of the Cumberland pencil museum

Experimentation in pencil production was widespread across Europe during the 16th and 17th centuries. Around 1560, an Italian couple named Simonio and Lyndiana Bernacotti made what are probably the first blueprints for the modern, wood-encased pencil. Their concept involved the hollowing out of a stick of juniper wood, and placing a stick of graphite inside. Shortly thereafter, a superior technique to that of the Bernacottis was adopted: two wooden halves were carved, a graphite stick inserted, and the halves then glued together — essentially the same method in use to this day.

The first attempt to manufacture graphite sticks from powdered graphite was in Nuremberg, Germany, in 1662, using a mixture of graphite together with the elements sulphur and antimony.

Today, pencils are manufactured around the world, using graphite from a number of sources. Most manufacturers use a mix of graphite and clay as the core of their pencils. Each of these substances is ground into a fine powder, which goes on to be cleaned in order to remove any impurities. After the two types of powders have been dried, they are mixed together using water. The amount of clay added to the mixture depends on how hard the pencil is intended to be — higher proportions of clay will make the pencil harder — and the amount of time spent grinding the mixture determines the quality of the core. The graphite-clay mixture is then shaped into long sticks, and after drying these are heated in a kiln to 1,200°C. In the next step in the process, the graphite-clay sticks are immersed in melted wax, which is absorbed into the mixture and helps to make it write more smoothly. In the next stage, the manufacturers use thin planks of cedar wood or juniper wood. Several long parallel grooves are cut into the planks, and the graphite-clay sticks are inserted into these. A second plank which has been prepared in the same way is then glued on top. The whole assembly is then cut into separate pencils. The outside may then be varnished, retaining the natural colour of the wood, or painted in different colours.

Questions 1–3

Complete the notes below.

Choose ***ONE WORD ONLY*** *from the passage for each answer.*

Write your answers in boxes 1–3 on your answer sheet.

In the 1500s, Borrowdale graphite

- was used by farmers in the area to indicate which **1** __________ they owned
- was first believed to be a type of **2** __________
- was encased in string to produce early pencils
- was recognised by the English **3** __________ as important in the manufacture of cannon balls

Questions 4–8

Do the following statements agree with the information given in Reading Passage 1?

In boxes 4–8 on your answer sheet, write

TRUE	*if the statement agrees with the information*
FALSE	*if the statement contradicts the information*
NOT GIVEN	*if there is no information on this*

4 Borrowdale graphite was superior to all other forms of graphite.

5 Pencil production in the Borrowdale area only lasted until the 1860s.

6 The Cumberland pencil museum contains examples of pencils from different European countries.

7 A technique which improved on the Bernacottis' method of making pencils is still used now.

8 In 1662, manufacturers in Nuremberg made pencils with graphite imported from England.

Questions 9–13

Complete the flow-chart below.

*Choose **ONE WORD ONLY** from the passage for each answer.*

Write your answers in boxes 9–13 on your answer sheet.

How pencils are manufactured today

Graphite and clay are separately crushed to form powders, then **9** __________ and dried
↓
the graphite and clay powders are combined with water
↓
the mixture is shaped into long sticks
↓
then, the sticks are dried, and **10** __________
↓
the sticks are dipped into **11** __________
↓
the sticks are put into **12** __________ in a wooden plank and another plank is fixed on top
↓
the whole thing is cut into individual pencils
↓
the pencils are **13** __________ or painted

Questions 1–3 Notes Completion

题号	答案	题干翻译	精确定位句	定位句翻译	详细解释
1	**sheep**	在1500年代，Borrowdale 石墨被当地农民用来标记他们拥有的哪些 ______。	“the locals found the mineral useful for marking their **sheep** so they could identify the ones that belonged to them.”	当地人发现这种矿物可以用来给他们的羊做标记，这样他们就能辨认哪些羊属于自己。	题干中的 **farmers in the area** 对应原文的 **locals**，**indicate which ... they owned** 对应 **identify the ones that belonged to them**。空格要求 ONE WORD ONLY，所以答案是 **sheep**。
2	**lead**	Borrowdale 石墨最初被认为是一种 ______。	“It was thought to be a form of **lead**, and so was referred to as plumbago...”	人们认为它是一种**铅**，所以称它为 plumbago。	题干的 **was first believed to be a type of** 对应原文 **was thought to be a form of**。注意后来它被认定为 carbon，但“最初被认为”是 **lead**。
3	**government**	Borrowdale 石墨被英国 ______ 认为在制造炮弹方面很重要。	“the **government** soon realised the value of the Borrowdale find, particularly for national defence purposes, because graphite proved ideal for lining the moulds used in the making of cannon balls.”	**政府**很快意识到 Borrowdale 发现物的价值，尤其是出于国防目的，因为石墨被证明非常适合用于制造炮弹时模具的内衬。	题干中的 **was recognised ... as important** 对应原文 **realised the value**；**in the manufacture of cannon balls** 对应 **used in the making of cannon balls**。所以答案是 **government**。

Questions 4–8 TRUE / FALSE / NOT GIVEN

题号	答案	题干翻译	精确定位句	定位句翻译	详细解释
4	**TRUE**	Borrowdale 石墨比所有其他形式的石墨都更优质。	“Although deposits of graphite existed in other parts of the world, they were not of the same exceptional purity and quality as Borrowdale graphite...”	虽然世界其他地方也有石墨矿藏，但它们并不具备 Borrowdale 石墨那样卓越的纯度和质量。	题干说 Borrowdale graphite **was superior to all other forms of graphite**，原文说其他地方的石墨没有 Borrowdale 石墨那样的 purity and quality，即 Borrowdale 更优质。两者一致，所以选 **TRUE**。
5	**FALSE**	Borrowdale 地区的铅笔生产只持续到19世纪60年代。	“English pencils continued to be made with sticks cut from pure Borrowdale graphite into the 1860s... The town of Keswick, close to the Cumberland graphite find, still manufactures pencils today...”	英国铅笔一直到19世纪60年代仍使用从纯 Borrowdale 石墨切割出的石墨条制造……靠近 Cumberland 石墨发现地的 Keswick 镇至今仍在生产铅笔。	题干说 **only lasted until the 1860s**，表示“只持续到19世纪60年代就停止”。但原文说 Keswick **still manufactures pencils today**，说明该地区附近今天仍生产铅笔。题干与原文矛盾，所以是 **FALSE**。
6	**NOT GIVEN**	Cumberland 铅笔博物馆收藏了来自不同欧洲国家的铅笔样品。	“the factory also being the location of the Cumberland pencil museum”	这家工厂也是 Cumberland 铅笔博物馆所在地。	原文只说这个工厂是博物馆所在地，没有提到博物馆里面是否有 **examples of pencils from different European countries**。这是典型的“有相关名词，但具体内容未提及”，所以是 **NOT GIVEN**。
7	**TRUE**	一种改进了 Bernacottis 制笔方法的技术现在仍在使用。	“Shortly thereafter, a superior technique to that of the Bernacottis was adopted... — essentially the same method in use to this day.”	不久之后，一种比 Bernacottis 方法更优越的技术被采用了……本质上这就是今天仍在使用的方法。	题干中的 **improved on the Bernacottis' method** 对应原文 **a superior technique to that of the Bernacottis**；**is still used now** 对应 **in use to this day**。信息完全一致，所以是 **TRUE**。
8	**NOT GIVEN**	1662年，Nuremberg 的制造商使用从英格兰进口的石墨制造铅笔。	“The first attempt to manufacture graphite sticks from powdered graphite was in Nuremberg, Germany, in 1662, using a mixture of graphite together with the elements sulphur and antimony.”	第一次尝试用石墨粉制造石墨条是在1662年的德国 Nuremberg，使用的是石墨、硫和锑的混合物。	原文确实提到 **Nuremberg, 1662, graphite**，但没有说石墨是否 **imported from England**。由于“来源于英格兰”这个信息没有出现，不能判断真伪，所以是 **NOT GIVEN**。

Questions 9–13 Flow-chart Completion

题号	答案	题干翻译	精确定位句	定位句翻译	详细解释
9	**cleaned**	石墨和黏土分别被压碎成粉末，然后被 ______ 并烘干。	"Each of these substances is ground into a fine powder, which goes on to be **cleaned** in order to remove any impurities."	每种物质都被磨成细粉，然后会被**清洁 / 净化**，以去除杂质。	题干中的 **crushed to form powders** 对应原文 **ground into a fine powder**。后面流程是 **cleaned**，再根据下一句 "After the two types of powders have been dried" 可知题干的 "and dried" 对应 dried。答案填 **cleaned**。
10	**heated**	然后，这些石墨黏土条被烘干，并被 ______。	"The graphite-clay mixture is then shaped into long sticks, and after drying these are **heated** in a kiln to 1,200°C."	石墨黏土混合物随后被塑造成长条，干燥之后，这些长条会在窑中被**加热**到1200摄氏度。	题干 **the sticks are dried, and...** 对应原文 **after drying these are heated**。空格需要过去分词，答案是 **heated**。
11	**wax**	这些长条被浸入 ______ 中。	"In the next step in the process, the graphite-clay sticks are immersed in melted **wax**..."	在接下来的步骤中，石墨黏土条被浸入融化的**蜡**中。	题干中的 **dipped into** 对应原文 **immersed in**，都是"浸入"。空格前是 into，答案直接填名词 **wax**。
12	**grooves**	这些长条被放入木板中的 ______，另一块木板被固定在上面。	"Several long parallel **grooves** are cut into the planks, and the graphite-clay sticks are inserted into these. A second plank... is then glued on top."	木板上会切出几条长长的平行**凹槽**，石墨黏土条被插入这些凹槽中。然后另一块准备好的木板被粘在上面。	题干 **put into ... in a wooden plank** 对应原文 **inserted into these**，而 these 指代前一句的 **grooves**。注意不能填 planks，因为条状物不是放进木板本身，而是放进木板里的凹槽。
13	**varnished**	铅笔被 ______ 或上漆。	"The outside may then be **varnished**, retaining the natural colour of the wood, or painted in different colours."	外部随后可以被上**清漆**，保留木材的自然颜色，或者被涂成不同颜色。	题干 **the pencils are ... or painted** 对应原文 **may then be varnished... or painted**。由于题干已给出 painted，所以空格填与 painted 并列的过去分词 **varnished**。

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